# Harshith Kadaru

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## EDUCATION

**The George Washington University** — Washington, DC
*Master of Science, Data Science, GPA: 4.0* — Expected May 2027

**CVR College of Engineering** — Hyderabad, India
*Bachelor of Technology, Computer Science and Information Technology* — April 2025

## SKILLS

**Languages:** Python, SQL, R, TypeScript, JavaScript
**ML and Statistics:** scikit-learn, LightGBM, PyTorch, pandas, SciPy, time series forecasting, A/B testing, power analysis
**AI and LLMs:** LLM evaluation, LLM-as-a-judge, agent harnesses, MCP servers, tool calling, RAG, embeddings, ChromaDB
**Engineering:** FastAPI, React, Redis, MySQL, MongoDB, Docker, GCP (Cloud Run, Cloud SQL), GitHub Actions CI

## EXPERIENCE

**MATClinics (SUDx Analytics)** — Maryland (Hybrid)
*Software Engineering Intern* — Jun 2026 - Aug 2026

- Migrated two production ML scoring pipelines from R / H2O / Java to Python, with 100% and 97.3% parity.
- Root-caused 5 months of silent job failures, backfilled the missing scores, and made the monthly rebuild idempotent.
- Moved real-time updates from in-memory WebSockets to Redis pub/sub to work across scaled Cloud Run instances.
- Built Python ETL into MySQL (Cloud SQL) and FastAPI telehealth services (Google Meet, Twilio SMS) with audit logging.
- Shipped 15+ React / TypeScript features to a multi-site clinical dashboard from direct provider feedback.

**The George Washington University** — Washington, DC
*Graduate Instructional Assistant, DATS 1001: Data Science for All* — Aug 2026 - Present

- Lead labs, grading, and office hours for R-based data wrangling, visualization, and statistical inference.

## PROJECTS

**claimcheck | Python, Docker, GitHub Actions | GitHub** — In Progress
*Open-source benchmark testing whether AI coding agents falsely claim success*

- Built 205 tasks with verified ground truth and a harness running GPT-OSS 120B and 20B through 530 runs at $0.
- Hardened grading against 8 ways to fake a pass; 12 cheat strategies (1,236 attempts) get none through.
- Built an LLM-as-a-judge from a different model family and pre-registered the analysis (McNemar and Fisher tests).
- Caught GPT-OSS 120B reverse-engineering test data to pass an impossible task, then quietly claiming success.

**claimcheck-mcp | Python, MCP, Docker | GitHub** — Oct 2026

- Built an MCP server giving AI agents 3 tools over claimcheck's 205 tasks (list, read, run in a Docker sandbox); found and fixed a symlink leak of answer files, with 96 tests in CI.

**A/B Experimentation and Decision Framework | Python, SciPy, pandas | GitHub** — Personal

- Built SRM checks, z-tests, Welch's t-tests, and guardrails that decide ship, continue, or rollback for each experiment.
- Showed with 2,000 A/A simulations that peeking raised false positives from 5% to 19%; O'Brien-Fleming restored 5%.
- Added power analysis (sample size, achieved power, test duration) validated against statsmodels, with 92 tests in CI.

**Demand Forecasting with Backtesting | Python, LightGBM, statsmodels | GitHub** — Oct 2026

- Forecast weekly sales for 3,049 Walmart (M5) items with Holt-Winters and LightGBM using lag and calendar features.
- 6-fold rolling-origin backtests with leakage tests: LightGBM MASE 0.963 vs naive 1.437; intervals hit 81% and 95%.

**RiskBot | Python, FastAPI, ChromaDB, GPT-OSS 120B | GitHub | Live Demo** — Dec 2025

- Flags sensitive HR and personal data across whole files, with document Q&A using top-k retrieval in ChromaDB.
- Cut peak memory from over 650 MB to 288 MB by swapping PyTorch for an ONNX embedder, to fit a 512 MB free tier.

## ACHIEVEMENTS

- **Winner, GW Data Science Association Datathon 2026:** built a 24-page Python and Streamlit platform analyzing 500K+ rows, using K-Means clustering, regression, and simulation to recommend policy interventions.